JOSEPH MOSKOVITZ

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PERSONAL STATEMENT

Aerospace Engineering BS/MS candidate specializing in electric propulsion and nuclear technology, with a strong foundation in M&A finance and space policy. Experienced in international and collaborative academic research across multiple international teams. Passionate about driving the long-term technological development and commercialization of nuclear power and emerging technologies within the space domain.

EDUCATION

GEORGIA INSTITUTE OF TECHNOLOGY — Atlanta, GA

Master's in Science in Aerospace Engineering — Aug 2026 - May 2027

Cumulative GPA: 4.0/4.0

Specialized Coursework: Heat Transfer, Radiation Transport, Nuclear Materials

Bachelor's in Science in Aerospace Engineering — Aug 2022 - May 2026

Minor in Nuclear & Radiological Engineering

Cumulative GPA: 4.0/4.0 | Highest Honors | Dean's List 2022-2026

Specialized Coursework: Reactor Physics, Plasma Physics & Fusion Engineering, Radiation Protection Engineering, Space Policy, Finite Element Analysis, Scientific Machine Learning

RANSOM EVERGLADES SCHOOL — Miami, FL

High School Diploma — Aug 2018 - May 2022

Cumulative GPA: 4.0/4.0 | Cum Laude

RESEARCH EXPERIENCE

HIGH-POWER ELECTRIC PROPULSION LAB - GEORGIA TECH — Atlanta, GA

Undergraduate Research Assistant — Jan 2024 – current

- **Machine Learning (ML) to Predict Low-Power HET Performance Operating on Multiple Propellants**
  - Collaborating with an international team (US, India, Israel) to develop a HET data preprocessing pipeline and evaluate various ML models predicting output parameters across various propellants (Xe, Kr, Ar, CO_2, N_2).
  - *Note: accepted to AIAA SciTech 2027 and under peer review at the Journal of Electric Propulsion*
- **Comprehensive Survey of Electric Propulsion (EP) Missions [2017-2025]**
  - Developed python-based web scrapers and automation scripts to extract and aggregate comprehensive mission data from ESA's DISCOS and Space-Track.org databases.
  - Analyzed orbital data to evaluate and quantify industry-wide adoption trends of EP technologies over the past decade.
- **Retarding Potential Analyzer (RPA) Redesign**
  - Redesigned the lab's proprietary RPA housing, grids, and wiring to more accurately analyze different plasmas and prevent shorting of different potential grids on older models.
  - Rebuilt CAD assemblies in SolidWorks, generating part drawings and GD&T with rigorous Design for Manufacturing (DFM) considerations for small, fragile components.
  - Created a BOM for the RPA, talked with various manufacturers and received quotes for different parts, machined parts in-house and ordered parts externally via lab management & procurement platform, and finally assembled 6+ fully functional units with interchangeable & spare parts.

EQUIPO DE PROPULSIÓN ESPACIAL Y PLASMAS - UC3M — Madrid, ES

Undergraduate Research Assistant — Sep 2024 – Dec 2024

- Developed a centralized machine/computer controls interface with LabVIEW for all power supply units (PSUs) involved in operating a HET.

ELECTRIC PROPULSION LABORATORY - TECHNION — Haifa, IL

Undergraduate Researcher — Jun 2024 – Aug 2024

- Characterized the performance and efficiency of the Simplified CAMILA low-power Hall-Effect Thruster (HET) across a diverse set of monatomic and molecular propellants (Xe, Kr, Ar, CO_2, N_2).
- Evaluated the viability of alternative propellants to mitigate industry reliance on Xe.
- *Note: Findings published at IEPC 2025 and at the Journal of Electric Propulsion.*

WORK EXPERIENCE

REALTA FUSION Madison, WI
Fusion Intern May 2026 – Aug 2026

- Developed a framework to cross-validate neutral beam injection (NBI) beam divergence from multiple existing diagnostics to better quantify beam characteristics.
- Designed and coordinated an experimental campaign with results to be presented at the American Physical Society Depart of Plasma Physics (APS DPP) conference in November.

ALLEN & COMPANY New York City, NY
Investment Banking Summer Analyst Jun 2025 – Jul 2025

- Executed live deals alongside senior bankers across the Biopharma, SaaS, and sports industrial verticals.
- Leveraged comprehensive valuation techniques (DCF, comps, precedents), M&A analysis (accretion/dilution), and rigorous equities research to support transaction advisory.

GEORGIA TECH COLLEGE OF ENGINEERING Atlanta, GA
Deformable Bodies Teaching Assistant Jan 2024 – May 2024

- Developed homework problems, facilitated office hours, and graded homework and exams for GT COE 3001, ensuring comprehension of complex engineering mechanics.

PUBLICATIONS

- J. Moskovitz, W. Brabston, D. Lev, M. L. R. Walker, M. Rubanovych, and V. Balabanov. "Performance Comparison and Analysis of Low Power Hall Thruster Operation on Inert and Molecular Propellants." 39th International Electric Propulsion Conference, September 14- 19 2025, London, UK. (IEPC-559)
- J. Moskovitz, W. Brabston, D. Lev, M. L. R. Walker, M. Rubanovych, and V. Balabanov. "Performance comparison and analysis of low-power hall thruster operation on atomic and molecular propellants." *J Electric Propulsion* **5**, 32 (2026). https://doi.org/10.1007/s44205-026-00199-5
- J. Moskovitz, A. Cooper, K. Shih, D. Sutherland, and C. Forest. "Beam-Divergence Spectroscopy on the WHAM Neutral Beam Injector using a Voltage-History Deconvolution Framework." 68th Annual Meeting of the APS Division of Plasma Physics, November 2–6 2026, Chicago, IL. (Poster presentation)
- S. Parmar, J. Moskovitz, D. Lev, and M. L. R. Walker. "Study of Machine Learning Techniques for Performance Prediction in Low-Power Hall Effect Thruster Operating on Monoatomic and Molecular Propellants." Research Square Preprint (2026). https://doi.org/10.21203/rs.3.rs-10583210/v1

PROJECTS

SCRAP-SAT Atlanta, GA
Magnetics Engineer & Finance Lead Jan 2026 – current

- Developing an integrated satellite to detumble and deorbit uncooperative space debris in Low-Earth Orbit (LEO) using magnetic eddy braking and a GOLD passive drag package. *Note: white paper accepted to AIAA SciTech 2027.*

1-D REACTION WHEEL Atlanta, GA
Project Lead Jan 2026 – May 2026

- Led the design and construction of a reaction/momentum wheel to demonstrate spacecraft attitude control systems.
- Modeled open and closed-loop system behavior in Simulink and designing a PID controller for precision attitude control.

COMMUNITY LEADERSHIP

CHABAD ON CAMPUS Atlanta, GA
Executive Board Member Jan 2024 – current

- Plan holidays, parties, social events, and more for the Jewish community at Georgia Tech.
- Increased the number of events for each Jewish holiday and helped increase weekly Shabbat attendance by over 50% in 3 years.

ZETA CHAPTER OF ALPHA EPSILON PI Atlanta, GA
Exchequer Apr 2023 – Apr 2025

- Managed the finances of the fraternity, including collecting dues, filing taxes, paying bills, and setting the budget.
- Incorporated the Chapter as a 501(c)(7) nonprofit organization, acquired a corporate line of credit, implemented modern expense-tracking software, and authored the organization's official Expense Policy and had it passed into the bylaws.

ADDITIONAL

Technical Skills: SolidWorks, Python, MATLAB, Simulink, LabVIEW, Finite Element Method Magnetics (FEMM), Data Analysis, Machine Learning, Excel, Woodworking, 3D Printing, Circuit design & construction
Languages: Bilingual in English & Spanish, Conversational in Hebrew